

Multi channel spectrum prediction algorithm based on GCN and LSTM

1st Han Zhang

College of Information and
Communication Engineering
Harbin Engineering University
Harbin, China
hldzh1997@hrbeu.edu.cn

2nd Qiao Tian

College of Computer Science
and Technology
Harbin Engineering University
Harbin, China
tianqiao@hrbeu.edu.cn

3rd Yu Han*

College of Information and
Communication Engineering
Harbin Engineering University
Harbin, China
yu.han1001@foxmail.com

Abstract—With the increasingly serious shortage of spectrum resources, spectrum dynamic access based on spectrum prediction technology is widely recognized. Due to the high burstiness and complex intrinsic correlation of spectrum monitoring data, high-precision multi-channel spectrum prediction is challenging. This paper constructs spectrum monitoring data as a kind of graph structure data based on the correlation of spectrum itself, and designs a graph network model combining Graph convolution network(GCN) and Long-short term memory network(LSTM) for multi-channel spectrum prediction. This paper creatively introduces the method of graph network. And GCN is used instead of CNN to extract the correlation of channels, so as to improve the accuracy of multi-channel prediction. Experiments are conducted based on a real-world spectrum measurement dataset. The results show that the model proposed in this paper has better predictive performance compared with other methods.

Index Terms—spectrum prediction, multivariate time series prediction, GCN, LSTM

I. INTRODUCTION

In recent decades, communication technology has experienced rapid development from 3G to 4G and then to 5G. In addition, in recent years, with the gradual maturity of artificial intelligence technology, more and more deep learning technologies have been applied to the field of communication [1] [2] [3] [4]. The update of these technologies makes more and more advanced communication equipment designed and widely used. More and more spectrum resources are allocated to different communication systems. This leads to the scarcity of unallocated spectrum resources. On the other hand, the low-frequency spectrum utilization of the existing static allocation strategy further aggravates the shortage of spectrum resources. How to dynamically use scarce spectrum resources has become the focus of researchers [5]. In order to dynamically and effectively use the idle spectrum resources, researchers have proposed the concept of dynamic spectrum access. That is, without introducing interference, the underutilized authorized frequency band is dynamically shared by a large number of unauthorized users to realize the reuse of spectrum resources. Accurate prediction of spectrum usage is the key premise to realize spectrum sharing and dynamic spectrum access. Spectrum prediction is a method to infer future channel occupancy status, duty cycle, power spectrum and other information based

on historical spectrum detection data. It is a widely accepted method to improve the overall utilization of spectrum by predicting spectrum holes and temporarily allocating them to unauthorized users. The existing research results show that the use of spectrum is highly sudden and highly correlated. Making full use of the time, frequency and spatial correlation obtained from historical observation data is the key to achieve high-precision spectrum prediction.

Early researchers used some traditional methods, mainly including methods based on statistics [6] and Markov models [7]. In recent years, tensor based methods have also attracted the attention of researchers [8]. With the development of deep learning, especially the wide application of RNN in the field of time series prediction, researchers have begun to study spectrum prediction methods based on deep learning. In [9], a LSTM model is used to predict the channel occupancy state and channel quality, and achieves better results than MLP. In [10], a 2D-LSTM-based spectral-temporal model is proposed to predict PSD values in different frequency ranges of the measured spectral dataset, which has better performance compared to baseline AR, Bayesian inference, and 1D-LSTM models. In [11], a model combining CNN and LSTM is designed, which uses CNN and LSTM to learn the correlation between frequency domain and time domain respectively. In [12], authors used a seq2seq model with attention mechanism for multi-step spectral prediction, and achieved good results on a simulated dataset.

Graph Neural Network (GNN) is an emerging network architecture in recent years [13]. Because more data representations in the real world are not a simple arrangement of sequences or planes, but more complex graph structures, graph networks have been widely used in some fields. In the field of traffic prediction, spatiotemporal graph networks have become the focus of research, and graph convolution (GCN)-based network architectures such as STGCN [14] and MTGNN [15] have been widely accepted. Inspired by the above studies, and considering the similar data structure and methodology of spectrum prediction and traffic prediction, this paper studies a graph neural network method for spectrum prediction. Existing deep learning methods for spectrum prediction often use convolution to learn the relationship between

multivariate variables (the relationship in frequency domain) to assist prediction. However, we believe that the relationship between frequency points is always very complex, which need a more deep convolution to learn. In some reference [16] [17], scholars have made some achievements in time domain, frequency domain and spatial domain. We believe that it is meaningful to make full use of these prior information. Therefore, based on the correlation of the spectrum, this paper constructs the spectral data as graph-structured data and uses GCN and LSTM to learn the frequency-domain and time-domain correlations respectively. We do experiments on an open-source dataset to verify the effectiveness of our method. The main contributions of this paper are summarized as follows:

- In this paper, based on spectral correlation, spectral data is constructed as graph-structured data and a graph network model is used for spectral prediction. As far as we know, this is the first time that graph networks have been introduced into the field of spectrum prediction.
- This paper constructs a spectrum prediction network model based on CNN and LSTM. The model uses GCN and LSTM to learn the correlation in frequency domain and time domain respectively.
- This paper uses real open-source spectrum monitoring data to verify the proposed method. The results show that our model has better performance than the combination of CNN and LSTM or other methods based on LSTM. This shows the great potential of graph networks in the field of spectrum prediction.

The rest of this paper is organized as follows. Section II describes the basic definition of graph and the method of constructing graph data. In section III, our model is introduced in detail. The experiment results and analysis are given in Section IV. Finally, we conclude the paper in Section V.

II. PROBLEM FORMULATION

The spectrum prediction problem can be regarded as a prediction problem of multivariate time series. Let $z_t \in R^N$ denote the spectrum monitoring results of N channels at time step t , where $z_t[i] \in R$ denote the value of the i th channel at time step t . Given a sequence of historical P time steps of observation of N channels, $X = \{z_{t_1}, z_{t_2}, \dots, z_{t_P}\}$. Our goal is to predict a Q -step sequence of future, $Y = \{z_{t_{P+1}}, z_{t_{P+2}}, \dots, z_{t_{P+Q}}\}$. In this paper, the spectrum monitoring data are characterized by graph method. Below, we give some basic definitions of graphs and our methods of constructing graph structure data.

Definition of graph: A graph is formulated as $G = (V, E)$, where V is the set of nodes, and E is the set of edges. In our problem, each monitored channel is defined as a node of the graph, and the correlation between channels is defined as an edge.

Definition of node neighborhood: Let $v, u \in V$ to denote a node, and $e = (v, u) \in E$ to denote an edge pointing from u to v . The neighborhood of the node v is defined as $N(v) = \{u \in V | (v, u) \in E\}$. In this paper, we consider that the correlation

between channels is undirected, that is, $(u, v) \in E = (v, u) \in E$. So the spectrum data we constructed is an undirected graph structure data.

Definition of adjacency matrix: The adjacency matrix is a mathematical representation of a graph, denoted as $A \in R^{N \times N}$, with $A_{ij} = c > 0$ if $(v_i, v_j) \in E$, and $A_{ij} = 0$ if $(v_i, v_j) \notin E$. In general, the value of c can be 1 or be defined as a variable indicating the correlation between nodes. For example, in the field of traffic prediction, c value is often defined as the reciprocal of the spatial distance between sensors. In this paper, spearman rank correlation coefficient is used to define the adjacency matrix of spectrum data, and its expression is as follows:

$$\rho = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}}, \quad (1)$$

where ρ represents spearman rank correlation coefficient of historical observation data of two channels, x and y are 2 different channels, x_i and y_i represent the i^{th} observation of the channel x and y . In order to adapt to the representation of the graph structure, we only consider the K nodes with the highest correlation with a certain node. So the adjacency matrix in this paper is defined as follows:

$$\begin{cases} A_{ij} = \rho_{ij} & \rho_{ij} \in \text{topk}(\rho_i) \\ A_{ij} = 0 & \text{other} \end{cases}. \quad (2)$$

Fig. 1 shows the process of constructing spectrum monitoring data as graph structure data.

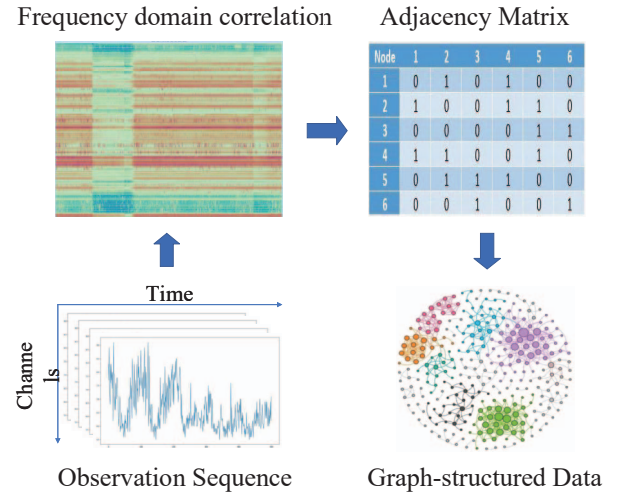


Fig. 1. Process of constructing spectrum monitoring data with graph structure.

III. PROPOSED MODEL

This paper presents a model of cascading GCN and LSTM. This model learns frequency domain correlation through GCN and time domain information through LSTM. Our complete prediction framework is shown in Fig. 2. The prediction model framework used in this paper is very similar to that in reference [11]. The main change is that we use GCN to replace CNN.

The main purpose of this paper is not to design a complex and perfect model, but to illustrate the great potential of graph network and graph convolution in the field of spectrum prediction by comparing this model with the popular method of combining CNN with LSTM or LSTM variants in recent years.



Fig. 2. The complete prediction framework.

A. Graph Convolution

Graph convolution is an extension of convolution operation in graph domain, and its core is spectral decomposition based on Laplace matrix. For graph $G = (V, E)$, its Laplace matrix can be defined as:

$$L = D^{-1/2} L D^{-1/2}, \quad (3)$$

where L is the Laplace matrix, D is the degree matrix of nodes, the elements on the diagonal are the degrees of each node in turn, and A is the adjacency matrix of the graph. This definition is also called symmetric normalized Laplacian, which is used in this paper and many reference. By definition, we can know that Laplace matrix is a positive semi definite symmetric matrix and has non-zero elements only at the central node and 1-hop neighbor node. Therefore, Laplace matrix can be decomposed:

$$L = U \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} U^T, \quad (4)$$

where $U = (\vec{u}_1, \vec{u}_2, \vec{u}_n)$ is the eigenvector matrix and λ is the eigenvalue. By comparing the Laplacian matrix of a graph with the Laplacian operator, the traditional fourier change $F(\omega) = \int f(t)e^{-i\omega t}dt$ can be extended to the graph:

$$\hat{f}(\lambda_l) = \sum_{i=1}^N f(i)u_l^*(i), \quad (5)$$

where f is the n -dimensional vector on the graph, $f(i)$ is the eigenvector of the node, and $u_l(i)$ represents the i -th component of the L -th eigenvector. Therefore, under the eigenvalue λ_l , the graph fourier transform of f is the inner product operation with the eigenvector u_l . Extended to the matrix form, the Fourier transform and inverse transform formulas of the graph are obtained:

$$\begin{aligned} \hat{f} &= U^T f \\ f &= U \hat{f} \end{aligned} \quad (6)$$

Compared with the traditional convolution formula $f * h = \mathcal{F}^{-1}[\hat{f}(\omega)\hat{h}(\omega)]$, the formula of graph convolution can be deduced as follows:

$$(f * h)_G = U \begin{pmatrix} \hat{h}(\lambda_1) & & \\ & \ddots & \\ & & \hat{h}(\lambda_n) \end{pmatrix} U^T f, \quad (7)$$

where $\hat{h}(\lambda_l) = \sum_{i=1}^N h(i)u_l^*(i)$ is the fourier transform of convolution kernel h on the graph. In general, the convolution kernel h can be designed as:

$$\hat{h}(\lambda_l) = \sum_{j=0}^K \alpha_j \lambda_l^j, \quad (8)$$

where $(\alpha_0, \alpha_1, \dots, \alpha_{K-1})$ is the learnable parameter in the model. This ingenious method avoids the problem of designing convolution kernel directly and can greatly reduce the complexity of parameters. Where K is the receptive field of the convolution kernel, that is, the weighted sum is performed in the k -order neighborhood of the central node. Fig. 3 intuitively shows the graph convolution when $K = 1$.

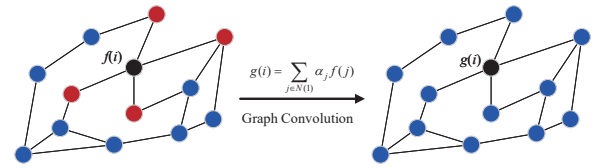


Fig. 3. Schematic diagram of graph convolution when $K = 1$

Generally speaking, the number of layers of GCN is not very deep. Because a large K value will make the receptive field too large. It means that the back propagation of the network will lead to over-smoothing, so that the characteristics of all nodes always look the same. In our model, we use a two-layer GCN structure to learn the frequency domain correlation.

B. LSTM Network

Long short term memory (LSTM) is a widely used time recursive network. LSTM is a gated RNN. The advantage of LSTM is to change the weight of self circulation by increasing input threshold, forgetting threshold and output threshold. When the model parameters are fixed, the integration scale can change dynamically at different times, thus avoiding the gradient disappearance or gradient explosion problem when the general RNN deals with long-term dependence (nodes far away in the time series). The LSTM cell is shown in the Fig. 4.

The main working principle of LSTM can be expressed by the following formula:

$$\begin{aligned} f_t &= \sigma(W_x^f x_t + W_h^f h_{t-1} + b_f) \\ i_t &= \sigma(W_x^i x_t + W_h^i h_{t-1} + b_i) \\ o_t &= \sigma(W_x^o x_t + W_h^o h_{t-1} + b_o) \\ C_t &= f_t * C_{t-1} + i_t * \tanh(W_x^C x_t + W_h^C h_{t-1} + b_C) \\ h_t &= o_t * \tanh(C_t) \end{aligned}, \quad (9)$$

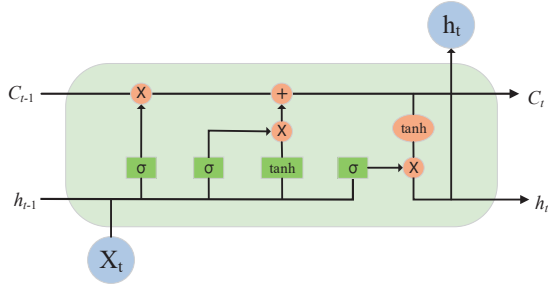


Fig. 4. A classic LSTM cell

where f_t , i_t and o_t are the outputs of forgetting gate, input gate and output gate respectively, C_t and h_t represent cell state and hidden state respectively, and W and b represent different weight matrices and bias vectors. In short, LSTM controls what to forget, remember and output at each time step by forgetting gate, input gate and output gate. This affects the unit state C_t , and ultimately achieves the purpose of long-term memory.

Many models based on LSTM are used in the field of spectrum prediction. Generally speaking, the model of seq2seq architecture is considered to be more effective in multi-channel prediction.

IV. EXPERIMENTS

A. Dataset

The data used in the experiment comes from the open source data platform of Electrosense [18]. We selected a sensor located around Madrid, Spain, and obtained PSD monitoring data in the 790-820MHz frequency band for about two months from May 2, 2021. The sensor is low-cost RTLSDR with omni-directional antennas which are deployed indoors. The time resolution of the acquired data is 15 minutes and the frequency resolution is 0.5MHz. So our experiment is a multi-step spectrum prediction with 59 channels. In the experiment, we use the observed data of the past 12 hours to predict the PSD value in the next 3 hours. From the perspective of data form, this is a problem that a matrix shaped as (48, 59) predicts the matrix shaped as (12, 59). We divide the data set into training set, verification set and test set according to the ratio of 7:1:2. In the process of constructing graph structure data, we set $k = 9$ based on the observation and experience of Spearman rank correlation coefficient between channels. In other words, each node of our graph data has 9 connected nodes.

B. Results and Analysis

In the experiment, our model consists of two layers of GCN and two layers of LSTM. The baseline model settings for comparison are as follows: A two-layer LSTM network, A network combining two-layer CNN with convolution kernel size of 3 and two-layer LSTM, and a seq2seq network in which both encoder and decoder are single-layer LSTM. In the experiment, the number of hidden layers of all LSTM layers

is set to 512 and the number of channels of GCN and CNN is set to 256. In all model experiments, we use curriculum learning strategies. In other words, at the beginning of training, only one time step is predicted and only one time step loss is calculated. As the number of iterations increases, we gradually increase the prediction length of the model, so that the model can gradually learn difficult tasks.

We use mean absolute error (MAE), mean absolute percentage error (MAPE) and root mean square error (RMSE) to evaluate the performance of the models. All experiments were repeated five times to ensure the reliability of the results.

The average results predicted over 12 time steps are shown in Table I. our model has relatively 6.06% lower MAE, 5.58% lower MAPE and 7.92% lower RMSE than the model based on CNN and LSTM.

TABLE I
BASELINE COMPARISON UNDER MULTI-STEP PREDICTION

Model	MAE	MAPE	RMSE
CNN+LSTM	0.9640±0.0811	8.0538±0.5424%	1.9619±0.2946
LSTM	1.0613±0.1148	8.7413±0.7953%	2.1358±0.3446
seq2seq LSTM	1.0087±0.1002	8.3143±0.6605%	2.0206±0.3205
GCN+LSTM	0.9056±0.0640	7.6047±0.4633%	1.8066±0.2367

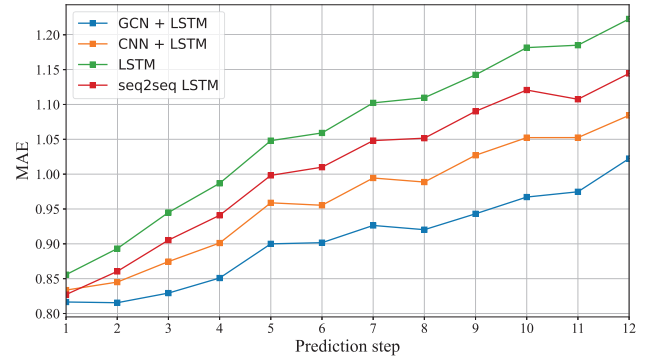


Fig. 5. MAE performance of the models under different time steps

Fig. 5 plots the MAE performances of our model and baseline at different time steps. Our model based on GCN and LSTM always achieves the best performance at any time step, followed by the method based on CNN and LSTM, and then the LSTM network based on seq2seq architecture, while the basic LSTM network has the worst effect. It can be observed that with the increase of prediction time steps, the advantages of our model become more obvious. This is because our method constructs a graph structure data with channel correlation implied, so the GCN based model has the best performance. The model based on CNN does not have prior information, so its performance is slightly weaker than our model. Other LSTM based models lack the corresponding frequency domain learning module, so their performance is the worst.

Fig. 6 plots the MAE performances of the model with different GCN layers. Obviously, the performance of 1-layer

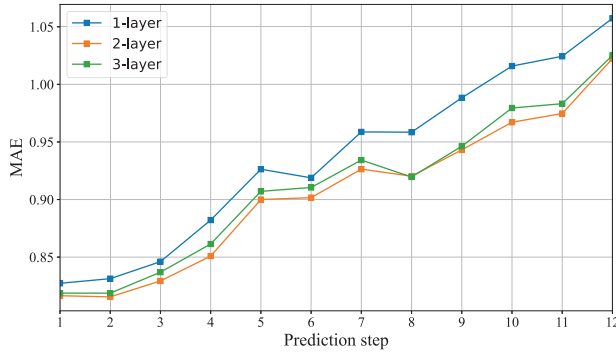


Fig. 6. MAE performance of different GCN layers

GCN is not as good as that of 2-layer and 3-layer models with more complex parameters. At the same time, an interesting phenomenon is worth noting: the performance of the model does not keep improving with the increase of the number of layers. The two-layer model achieves the best performance, while the performance of the three-layer model decreases slightly. This is because too deep GCN layer will lead to too large receptive field, which lead to the phenomenon of over-smooth, that is, all nodes always look similar after weighted average. This makes the node features lose diversity and leads to the degradation of model performance. This phenomenon limits the use of GCNs to some extent because it is difficult to design a very deep GCN network like CNNs. This is our next research direction.

V. CONCLUSIONS

In this paper, we propose a method of constructing spectrum structure data and a model based on GCN and LSTM for multi-channel and multi-step spectrum prediction tasks. We construct spectral structure data based on the spectral channel correlation represented by the Spearman rank correlation coefficient, and then extract the frequency domain and time domain information using GCN and LSTM respectively. Compared with CNN and other methods, our model has better ability to learn the correlation of channels, because the graph structure data we construct contains some prior knowledge. The experimental results show that our method is more effective than other models based on LSTM. It shows the great application prospect of graph network in the field of spectrum prediction. In future research, we will evaluate more graph construction methods and try deeper and more advanced graph models.

ACKNOWLEDGMENT

This work is supported by the Fundamental Research Funds for the Central Universities (3072022CF0601, 3072022CF0804).

REFERENCES

[1] Y. Tu, Y. Lin, J. Wang, and J.-U. Kim, "Semi-supervised learning with generative adversarial networks on digital signal modulation classification," *Comput. Mater. Continua*, vol. 55, no. 2, pp. 243–254, 2018.

[2] Y. Dong, X. Jiang, H. Zhou, Y. Lin, and Q. Shi, "Sr2cnn: Zero-shot learning for signal recognition," *IEEE Transactions on Signal Processing*, vol. 69, pp. 2316–2329, 2021.

[3] Y. Lin, Y. Tu, Z. Dou, L. Chen, and S. Mao, "Contour stella image and deep learning for signal recognition in the physical layer," *IEEE Transactions on Cognitive Communications and Networking*, vol. 7, no. 1, pp. 34–46, 2020.

[4] T. Ya, L. Yun, Z. Haoran, J. ZHANG, W. Yu, G. Guan, and M. Shiwen, "Large-scale real-world radio signal recognition with deep learning," *Chinese Journal of Aeronautics*, 2021.

[5] Y. Lin, M. Wang, X. Zhou, G. Ding, and S. Mao, "Dynamic spectrum interaction of uav flight formation communication with priority: A deep reinforcement learning approach," *IEEE Transactions on Cognitive Communications and Networking*, vol. 6, no. 3, pp. 892–903, 2020.

[6] Y. Chen and H.-S. Oh, "A survey of measurement-based spectrum occupancy modeling for cognitive radios," *IEEE Communications Surveys & Tutorials*, vol. 18, no. 1, pp. 848–859, 2014.

[7] X. Chen, H. Zhang, A. B. MacKenzie, and M. Matnimmikko, "Predicting spectrum occupancies using a non-stationary hidden markov model," *IEEE wireless communications letters*, vol. 3, no. 4, pp. 333–336, 2014.

[8] X. Li, Z. Liu, Y. Xu, X. Wang, and T. Song, "Recovering missing values from corrupted historical spectrum observations for dependable spectrum prediction," in *2020 IEEE/CIC International Conference on Communications in China (ICCC)*. IEEE, 2020, pp. 941–946.

[9] L. Yu, J. Chen, G. Ding, Y. Tu, J. Yang, and J. Sun, "Spectrum prediction based on taguchi method in deep learning with long short-term memory," *IEEE Access*, vol. 6, pp. 45 923–45 933, 2018.

[10] M. A. Aygöl, M. Nazzal, M. İ. Sağlam, D. B. da Costa, H. F. Ateş, and H. Arslan, "Efficient spectrum occupancy prediction exploiting multi-dimensional correlations through composite 2d-lstm models," *Sensors*, vol. 21, no. 1, p. 135, 2020.

[11] L. Zhang and M. Jia, "Accurate spectrum prediction based on joint lstm with cnn toward spectrum sharing," in *2021 IEEE Global Communications Conference (GLOBECOM)*. IEEE, 2021, pp. 1–6.

[12] Y. Gao, C. Zhao, and N. Fu, "Joint multi-channel multi-step spectrum prediction algorithm," in *2021 IEEE 94th Vehicular Technology Conference (VTC2021-Fall)*. IEEE, 2021, pp. 1–5.

[13] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and S. Y. Philip, "A comprehensive survey on graph neural networks," *IEEE transactions on neural networks and learning systems*, vol. 32, no. 1, pp. 4–24, 2020.

[14] B. Yu, H. Yin, and Z. Zhu, "Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting," in *Proceedings of the 27th International Joint Conference on Artificial Intelligence (IJCAI)*, 2018, pp. 3634–3640.

[15] Z. Wu, S. Pan, G. Long, J. Jiang, X. Chang, and C. Zhang, "Connecting the dots: Multivariate time series forecasting with graph neural networks," in *Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining*, 2020, pp. 753–763.

[16] A. Palaos, J. Riihijärvi, O. Holland, and P. Mähönen, "Detailed measurement study of spatial similarity in spectrum use in dense urban environments," *IEEE Transactions on Vehicular Technology*, vol. 66, no. 10, pp. 8951–8963, 2017.

[17] J. Sun, L. Yu, J. Li, and G. Ding, "Similarity analysis on spectrum state evolutions," in *International Conference in Communications, Signal Processing, and Systems*. Springer, 2018, pp. 502–510.

[18] S. Rajendran, R. Calvo-Palomino, M. Fuchs, B. Van den Bergh, H. Corobés, D. Giustiniano, S. Pollin, and V. Lenders, "Electrosense: Open and big spectrum data," *IEEE Communications Magazine*, vol. 56, no. 1, pp. 210–217, 2017.